Limitations of Low-Code Platforms for a Complex Enterprise Use Case

Aim

To analyze the technical, architectural, and operational limitations of low-code development platforms using the Zoho Creator Job Application Management application as a case study.

Theory

Low-code platforms such as Zoho Creator simplify application development through visual form builders, reports, workflows, and scripting tools. They are suitable for business applications such as recruitment management, employee management, and data tracking. However, complex enterprise requirements may introduce limitations in customization, performance, integration, and long-term maintenance.

1. Customization Constraints

Concept: Low-code platforms provide predefined components and visual development tools. Complex business logic may require additional scripting or external services.

Application: In the Job Application Management application, basic candidate registration, application tracking, interview scheduling, and approval workflows can be created easily. However, advanced candidate-ranking algorithms or complex recruitment analytics may require custom Deluge scripts or external applications.

Limitation: Highly customized features may be difficult to implement using only built-in components.

2. Scalability and Performance Limitations

Concept: As the number of records and users increases, applications may require additional optimization to maintain performance.

Application: The recruitment application stores candidate details, application statuses, interview information, and hiring records. A large number of applications may increase report-loading times and the processing requirements of workflows.

Limitation: Performance may depend on platform limits, data volume, query design, and the complexity of workflows.

3. Vendor Lock-in

Concept: Low-code applications often depend on platform-specific tools, data structures, and scripting languages.

Application: The recruitment application uses Zoho Creator forms, reports, Deluge scripts, and built-in workflow features. Moving the complete application to another platform may require rebuilding these components.

Limitation: Migration to another platform may require additional time, effort, and cost.

4. Complex Integration Challenges

Concept: Integrating low-code applications with external software may require APIs, authentication, connections, and additional error handling.

Application: The recruitment application can use API integration to retrieve information from external services. For example, the API integration experiment used the JSONPlaceholder service to retrieve sample user information through Deluge's invokeurl function.

Limitation: Integrating with real recruitment systems, external HR software, or large enterprise databases may require additional configuration and security measures.

Case Study Scenario: Enterprise Recruitment Management System

Enterprise Requirement:

A large multinational company wants to develop a recruitment management system that handles thousands of candidate applications, multiple interview rounds, department-wise

recruitment reports, automated notifications, and approval workflows.

Platform Chosen:

Zoho Creator is selected because it provides visual application development, form and report creation, workflow automation, and integration capabilities.

Identified Limitations During Implementation

1. Customization Challenges

   The existing application supports candidate registration, application tracking, and interview management. However, advanced candidate-ranking algorithms and complex recruitment analytics may require additional scripting or external AI services.

2. Performance Challenges

   As the number of candidate records increases, reports and workflows may require optimized queries, efficient data structures, and careful management of repeated database operations.

3. Integration Challenges

   The application can connect to external services using APIs. However, integration with enterprise HR systems may require authentication, error handling, data mapping, and connection configuration.

4. Platform Dependency

   Forms, reports, workflows, and Deluge functions are designed around Zoho Creator. Migrating the complete application to another platform may require redesigning these components.

5. Cost and Resource Considerations

   As the application grows, the organization must evaluate its Zoho Creator subscription, user requirements, usage limits, and integration needs. The overall cost depends on the selected plan and actual usage.

Result

The limitations of low-code platforms were analyzed using the Zoho Creator Job Application Management application as a case study. The analysis identified potential challenges in advanced customization, scalability, external integration, platform dependency, and cost management. It demonstrated that Zoho Creator is suitable for recruitment workflows and business applications, while complex enterprise requirements may need additional scripting, external services, or integration with traditionally developed systems.